# RAJU YELLATURI

**Sr. Gen AI/ML engineer**

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## PROFESSIONAL SUMMARY

- **Senior AI/ML & Generative AI Engineer with 9+ years of experience** designing, developing, and deploying enterprise-scale intelligent systems across finance, logistics, automotive, and consulting domains.
- Proven track record of delivering **production-grade Machine Learning and AI solutions** that solve complex business problems and drive measurable operational impact in high-volume environments.
- Extensive experience building **end-to-end ML pipelines**, including data ingestion, preprocessing, feature engineering, distributed model training, evaluation, optimization, deployment, and continuous monitoring.
- Strong expertise in **predictive modeling, fraud detection, anomaly detection, route optimization, time-series forecasting, and connected vehicle analytics**.
- Hands-on experience with advanced ensemble models such as **XGBoost, Random Forest, and Gradient Boosting** for high-accuracy prediction systems.
- Specialized in **Generative AI and Large Language Models (LLMs)**, including implementation of **Retrieval-Augmented Generation (RAG), embedding-based semantic search, and enterprise document intelligence platforms**.
- Experienced in designing scalable AI architectures using **vector databases, grounding techniques, and response validation frameworks** to enhance model accuracy and reduce hallucinations.
- Strong background in **distributed data processing and large-scale ML training using Apache Spark and PySpark** for structured and streaming datasets.
- Skilled in developing **real-time inference APIs** and integrating predictive models into enterprise systems for dynamic decision-making and automation.
- Well-versed in **MLOps best practices**, including MLflow-based model tracking, versioning, automated retraining, performance monitoring, and drift detection.
- Proven ability to productionize AI systems on **AWS and Microsoft Azure**, leveraging **Docker and Kubernetes** for scalable and secure cloud-native deployments.
- Strong understanding of **Responsible AI, governance frameworks, model explainability, RBAC, data privacy, and regulatory compliance**, particularly in risk-sensitive financial environments.
- Adept at collaborating with cross-functional teams to translate complex analytical challenges into **scalable, secure, and business-aligned AI solutions**.
- Passionate about building **reliable, interpretable, and production-ready AI systems** that deliver long-term business value and operational excellence.

## TECHNICAL SKILLS:

| Category | Skills & Technologies |
|---|---|
| **Programming & Data Processing** | Python, SQL, Pandas, NumPy, SciPy, Data Cleaning, Data Transformation, Exploratory Data Analysis (EDA) |
| **Machine Learning** | Supervised & Unsupervised Learning, Regression, Classification, Clustering, Ensemble Methods, Logistic Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM, K-Means, Feature Engineering, Model Evaluation (AUC, Precision, Recall, F1-Score), Cross-Validation, Hyperparameter Tuning |
| **Deep Learning** | Neural Networks, Feedforward Networks, CNN Fundamentals, RNN Concepts, Model Optimization Techniques |
| **Generative AI & LLMs** | Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Window Optimization, Response Evaluation, LangChain, LlamaIndex, Azure OpenAI, OpenAI APIs, LLM Grounding Techniques |
| **Embeddings & Vector Search** | Embedding Models, Sentence Transformers (concepts), Semantic Search, Vector Similarity Search, FAISS, Pinecone, Azure AI Search, Metadata Filtering |
| **Natural Language Processing (NLP)** | Text Preprocessing, Tokenization, Named Entity Recognition (NER) Concepts, Text Classification, Sentiment Analysis, Document Parsing, Information Extraction |
| **Big Data & Distributed ML** | Apache Spark, PySpark, Hadoop, Distributed Model Training, Batch Processing, Near Real-Time Processing |
| **MLOps & Model Lifecycle** | MLflow, Model Versioning, Model Registry, Model Monitoring, Data Drift Detection, Concept Drift Monitoring, Automated Retraining, CI/CD for ML, Performance Logging |
| **Cloud & AI Deployment** | AWS (EC2, S3, EMR, IAM), Microsoft Azure (Azure OpenAI, AKS, Azure Blob Storage), Docker, Kubernetes, REST APIs for Model Inference |
| **Streaming & Data Pipelines** | Apache Kafka, Streaming Data Processing, Batch & Real-Time Scoring Pipelines |
| **AI Governance & Security** | Responsible AI, Model Explainability (Conceptual), RBAC, Secure API Design, Data Privacy, PII Handling, Audit Logging, Compliance & Governance Standards |

## EDUCATION

- **Bachelors** in Electronics and communicational Engineering

## CERTIFICATIONS

- CompTIA Security+ - issued Apr 2023
- AWS Cloud Practitioner Essentials offered by AWS.

## PROFESSIONAL EXPERIENCE:

**PWC, Tampa, FL** **Nov 2024 – present**

**Sr. Gen AI & Python Developer**

**Project overview:**

Led the design and implementation of an enterprise-scale AI-driven Risk & Compliance Intelligence platform to automate the analysis of financial, regulatory, and policy documents for consulting and audit teams. The solution leveraged Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and machine learning-based risk classification models to deliver contextual, accurate, and explainable insights. Built scalable document ingestion and embedding pipelines, integrated semantic search using vector databases, and deployed secure AI microservices on Azure cloud infrastructure. Implemented MLOps, model governance, and Responsible AI frameworks to ensure compliance, monitoring, and production reliability, resulting in significant improvements in document review efficiency and proactive risk identification.

**Responsibilities:**

- Designed and developed an enterprise-scale **AI-powered Risk & Compliance Intelligence platform** to automate analysis of financial, regulatory, and policy documents.
- Built end-to-end **Retrieval-Augmented Generation (RAG) pipelines** integrating Large Language Models (LLMs) with enterprise-approved data sources to improve contextual accuracy and reduce hallucinations.
- Implemented scalable **document ingestion, preprocessing, chunking, and embedding generation workflows** for structured and unstructured datasets (PDF, DOCX, policy reports).
- Developed and optimized **semantic search architecture using vector databases (Azure AI Search, FAISS, Pinecone)** for high-performance document retrieval.
- Engineered **machine learning classification models** (XGBoost, ensemble methods) to predict and score compliance risk levels across financial documents.
- Applied advanced **feature engineering and model evaluation techniques (AUC, Precision, Recall, F1-score)** to optimize predictive performance.
- Designed secure **LLM integration using Azure OpenAI APIs** within a multi-tenant cloud-native architecture.
- Implemented **MLOps best practices**, including MLflow-based model tracking, versioning, automated retraining workflows, and data drift detection.
- Developed **real-time inference APIs using FastAPI**, enabling secure AI service consumption by internal consulting and audit teams.
- Containerized AI services using Docker and deployed scalable microservices on **Azure Kubernetes (AKS)**.
- Applied **Responsible AI principles, RBAC, data privacy controls, and audit logging frameworks** to ensure governance and compliance alignment.
- Improved document review efficiency by ~60% and enhanced proactive risk identification through AI-driven automation.

**Environment:** Python, SQL, scikit-learn, XGBoost, Ensemble Models, Feature Engineering, Model Evaluation (AUC, Precision, Recall, F1-Score), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Azure OpenAI, OpenAI APIs, Embeddings, Semantic Search, Azure AI Search, FAISS, Pinecone, NLP, Document Chunking, PDF/DOCX Parsing, MLflow, Model Monitoring, Data Drift Detection, Automated Retraining, CI/CD, FastAPI, REST APIs, Docker, Kubernetes (AKS), Microsoft Azure, RBAC, Responsible AI, Data Privacy, Audit Logging.

**Ups, Atlanta, GA** **Jan 2022 – Oct 2024**

**Sr. Machine Learning Engineer**

**Project overview:**

Designed and deployed a scalable machine learning platform to improve last-mile delivery efficiency through dynamic ETA prediction and route optimization. Built distributed ML pipelines to process high-volume GPS, traffic, weather, and operational datasets, integrating predictive models with real-time decision systems. Implemented automated retraining, monitoring, and cloud-native deployment to ensure accuracy, scalability, and long-term model reliability, resulting in improved delivery time precision and operational efficiency.

**Responsibilities:**

- Designed and implemented advanced **ETA prediction models** using **XGBoost and Gradient Boosting** to improve delivery time accuracy across large-scale logistics networks.
- Built **end-to-end machine learning pipelines** for data ingestion, feature engineering, distributed training, and evaluation using **Python and PySpark**.
- Engineered high-impact features from **GPS, traffic, weather, package volume, and driver behavior data** to enhance predictive accuracy.
- Leveraged **Apache Spark** for large-scale distributed processing of historical and near real-time logistics datasets.
- Developed **route optimization models** integrating ML predictions with operational constraints to reduce delivery delays and fuel consumption.
- Implemented **real-time inference APIs** enabling dynamic ETA updates and intelligent routing decisions.
- Applied advanced **model evaluation metrics (RMSE, MAE, AUC, F1-score)** and performed **hyperparameter tuning** to optimize performance and minimize prediction errors.
- Conducted **feature importance analysis and model explainability techniques** to improve transparency and operational trust in ETA predictions.
- Designed **automated retraining workflows** to adapt models to seasonal delivery trends and evolving traffic patterns.
- Implemented **model monitoring, performance tracking, and drift detection frameworks** to ensure long-term stability.
- Deployed ML workflows on **AWS cloud infrastructure** using **Docker and Kubernetes** for scalable, production-grade services.
- Optimized inference latency and system throughput to support **high-volume real-time logistics operations**.
- Improved delivery time accuracy and operational efficiency through **AI-driven automation and predictive optimization**.
- Developed and validated **ensemble-based predictive models** to dynamically adjust delivery schedules based on real-time traffic congestion, weather disruptions, and operational constraints.
- Established **model performance benchmarking frameworks** to continuously evaluate prediction stability across diverse geographic regions and peak delivery periods.

**Environment:** Python, SQL, scikit-learn, XGBoost, Gradient Boosting, Regression Models, Feature Engineering, Model Evaluation (RMSE, MAE, AUC, F1-Score), Apache Spark, PySpark, Distributed ML Training, AWS (EC2, S3, EMR), Docker, Kubernetes, REST APIs, Batch & Real-Time Inference Pipelines, Model Monitoring, Drift Detection, Automated Retraining, CI/CD.

**Ford motors, Palo Alto, CA** **Jan 2021 – Dec 2021**

**Senior Python developer**

**Project overview:**

Developed a machine learning-based Predictive Maintenance platform leveraging high-volume vehicle telematics data to detect component failures and optimize service scheduling. Built scalable ML pipelines with distributed processing, integrated real-time inference APIs, and deployed containerized services on AWS. Implemented monitoring and automated retraining to ensure long-term model reliability, resulting in reduced vehicle downtime and improved operational efficiency.

**Responsibilities:**

- Designed and developed machine learning models to predict vehicle component failures using high-volume telemetry data (engine diagnostics, battery health, braking systems, temperature signals).
- Built end-to-end ML pipelines for data ingestion, preprocessing, feature engineering, model training, validation, and deployment using Python.
- Engineered advanced time-series and statistical features from high-frequency sensor streams to improve early failure detection accuracy.
- Applied supervised learning techniques (Random Forest, XGBoost, Gradient Boosting) and anomaly detection models for predictive maintenance use cases.
- Developed time-window aggregation strategies to capture vehicle performance trends and degradation patterns.
- Leveraged Apache Spark and PySpark for distributed processing and scalable model training on large-scale telematics datasets.
- Implemented model evaluation frameworks using AUC, RMSE, Precision-Recall, and cross-validation to optimize performance and minimize false positives.
- Designed real-time inference APIs to deliver predictive insights into fleet dashboards and connected vehicle applications.
- Integrated ML predictions with operational systems to support proactive maintenance scheduling and reduce unexpected vehicle downtime.
- Built automated retraining workflows to continuously update models based on new sensor data and seasonal driving behavior patterns.
- Implemented model monitoring, performance tracking, and drift detection mechanisms to ensure long-term prediction stability.
- Containerized ML services using Docker and deployed scalable microservices on Kubernetes within AWS cloud infrastructure.
- Collaborated with data engineering teams to design scalable data architectures for continuous ingestion of telematics streams.
- Applied secure deployment practices, including RBAC and encrypted data handling, to ensure compliance with enterprise security standards.
- Improved maintenance prediction accuracy and reduced unplanned service incidents through AI-driven optimization and real-time analytics.

**Environment:** Python, SQL, scikit-learn, XGBoost, Random Forest, Gradient Boosting, Anomaly Detection, Time-Series Feature Engineering, Apache Spark, PySpark, AWS (EC2, S3, EMR), Docker, Kubernetes, REST APIs, Model Monitoring, Drift Detection, Automated Retraining, CI/CD, RBAC.

**Wells Fargo, charlotte, NC** **Nov 2019 – Dec 2020**

**python developer & ML engineer**

**Project overview:**

Developed and deployed machine learning models to detect fraudulent banking transactions using behavioral analytics and anomaly detection techniques. Built distributed ML pipelines and near real-time scoring systems, implemented automated retraining and drift monitoring, and ensured regulatory compliance through model governance frameworks. Improved fraud detection accuracy while reducing false positives and operational risk.

**Responsibilities:**

- Designed and deployed advanced supervised learning models (XGBoost, Random Forest, Logistic Regression) to classify fraudulent transactions with high predictive accuracy.
- Implemented anomaly detection techniques to identify emerging fraud patterns and unusual customer behavior.
- Engineered high-impact behavioral features including transaction velocity, spending deviation, geolocation anomalies, device fingerprinting, and customer risk profiles.
- Built distributed ML training pipelines using Python and PySpark to process high-volume historical transaction datasets.
- Applied model optimization techniques including hyperparameter tuning, cross-validation, and threshold calibration to balance precision-recall tradeoffs.
- Evaluated models using AUC-ROC, F1-score, Precision-Recall curves, and confusion matrix analysis to minimize false positives.
- Developed near real-time fraud scoring APIs to support dynamic risk-based transaction decisioning.
- Implemented automated model retraining workflows to adapt to evolving fraud tactics and seasonal transaction behaviors.
- Established model monitoring and data drift detection frameworks to ensure long-term stability and regulatory compliance.
- Maintained MLflow-based model tracking and versioning for governance and audit readiness.
- Collaborated with risk and compliance teams to align ML outputs with enterprise risk appetite and regulatory requirements.
- Improved fraud detection sensitivity while reducing operational false-positive investigations through AI-driven optimization.

**Environment:** Python, SQL, scikit-learn, XGBoost, Random Forest, Logistic Regression, Anomaly Detection, Feature Engineering, Hyperparameter Tuning, AUC-ROC, Precision-Recall Optimization, Apache Spark, PySpark, MLflow, REST APIs, Real-Time Scoring Pipelines, Model Monitoring, Drift Detection, CI/CD, Risk-Based Thresholding.

**Ninja cart, Hyderabad, India** **Jan 2018 – Sep 2019**

**python developer**

**Project overview:**

Developed Python backend services to manage order processing, inventory tracking, and fulfillment workflows within a supply-chain platform. Built REST APIs to integrate retailer and logistics systems, implemented validation logic to ensure data accuracy, and optimized database performance to handle high transaction volumes while maintaining system reliability.

**Responsibilities:**

- Developed scalable **Python backend services** to manage order processing, inventory updates, and end-to-end fulfillment workflows.
- Built and maintained secure **REST APIs (Django / Flask)** to integrate retailer, supplier, and logistics systems.
- Implemented data processing and aggregation pipelines to support **operational reporting and supply-chain analytics**.
- Designed and optimized **PostgreSQL/MySQL database schemas and queries** to handle high transaction volumes during peak periods.
- Enforced business validation rules to ensure **inventory accuracy and order consistency**.
- Integrated real-time logistics and delivery tracking systems to enable **shipment visibility**.
- Implemented logging, monitoring, and error-handling mechanisms to improve **system reliability and performance**.
- Collaborated with product and operations teams in an **Agile environment** to deliver scalable and business-aligned solutions.

**Environment:** Python, Django, Flask, REST APIs, PostgreSQL, MySQL, Celery, Redis, RabbitMQ, Pandas, NumPy, AWS (EC2, S3), Docker, Git, ELK Stack, Agile/Scrum

**Appulse technologies private limited, Coimbatore, India** **Oct 2016 – Dec 2017**

**Jr software developer**

**Project overview:**

Assisted in developing a web-based inventory and order management system by supporting backend modules, database updates, and UI enhancements using HTML, CSS, and JavaScript. Contributed to testing, defect fixes, and overall application improvements in a collaborative team environment.

**Responsibilities:**

- Contributed to the development of a **web-based inventory and order management system** for a retail client using ASP.NET MVC and C#.
- Assisted in building backend modules to manage product, order, and inventory data within a relational database environment.
- Implemented validation logic and supported **SQL Server database design and maintenance** to ensure data accuracy.
- Developed and enhanced UI components using **HTML, CSS, JavaScript, and Bootstrap**.
- Performed unit testing, defect resolution, and application enhancements to improve overall system functionality.
- Collaborated with senior developers in an **Agile team environment**, participating in daily updates and code management using Git.

**Environment:** ASP.NET MVC, C#, HTML, CSS, JavaScript, Bootstrap, SQL Server, Visual Studio, Git, Manual Testing, MVC Architecture